In-Memory Object Storage in a Single Address Space

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May 4, 2026

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Outline

Motivation

Approach

Benchmarks

Motivation

What is an in-memory object store?

- A shared key→value table holding live application objects.
- Multiple clients (threads or processes) read and write concurrently.
- Latency profile ranging from nanoseconds to milliseconds.
- Used as a cache, a feature store, or a coordination layer between tightly coupled services.
  - Often accessed remotely, but not always (e.g., data pipelines, multi-GPU training).
  - Focus today is on low-latency intra-server IPC.

In-Memory Bottlenecks

Can classify (most) overhead into one of two categories:

IPC overhead

- Context-switch on every `get`/`put`:
  - Notify database with `send`/`recv`
  - Schedule database worker thread
- Copying values from client memory to server memory and back

Metadata overhead

- Synchronization on shared state
  - Concurrent hash tables, skip lists
  - Key/value reference counts
- Memory access isn't free!
  - Data cache and TLB misses

Related Work

- "Rearchitecting In-Memory Object Stores for Low Latency" (VLDB '22).
- Elegant solution to the IPC problem:
  - Metadata (hash table) and data (raw buffers) allocated from a shared memory region.
  - Clients read/write to database directly via specialized client library.
- Unsurprisingly, benchmarks are far better than competitors like Redis and Plasma.[1]
- But... focus is placed on safety rather than "low latency":
  - "To avoid concurrent modification to the object store's state, we use a single spinlock to serialize client operations... Lightning has the best performance when there is only one client because multiple clients can result in lock contention."
  - Emphasis on fault isolation (formally verified!).

[1] deprecated Apache Arrow project

Towards a Single Address Space

- Suppose that clients never crash.
- Give up on the multi-process model:
  - Shared memory is often slower (due to TLB pressure) and harder to deal with (must use relative offsets instead of raw pointers) than local heap memory.
  - Ideally, a client can `put` an arbitrary pointer (e.g., to a graph or sparse matrix) and it is immediately accessible to other clients, like in multi-threaded applications.
  - Lose process isolation, but this is (partially) fixable.
- Use faster, harder-to-verify data structures like concurrent hash tables.
- But maybe it's impractical to refactor your codebase to use threads!

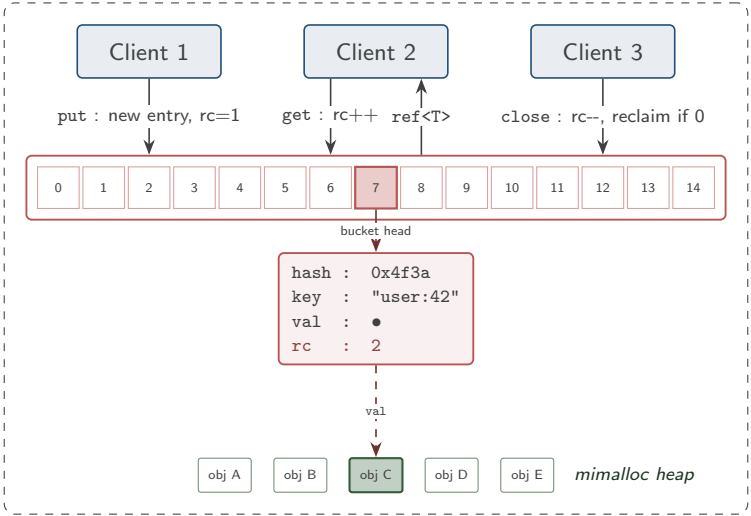
These motivate a design where *client binaries are loaded into a single process at runtime*.

Approach

System API

- Clients `include` a header with database operations, resolved at runtime to host code:
  - `put(key, value*, dtor)`: insert pointer to value and register destructor callback.
  - `get(key) -> handle`: retrieve ref-counted pointer to value.
  - `close(handle)`: decrement ref-count, call `dtor` and free when it reaches zero.
- Also some convenience functions, like a C++-only RAII helper for handles.
- Reference counts make the API flexible but are a real bottleneck at scale.

System Architecture



Concurrent Hash Tables

- For the database backend, prioritize throughput and tail latency at high concurrency.
- Initially tried `boost::concurrent_flat_map`:
  - Flat open-addressing scheme that uses SIMD for hash lookups.
  - Striped read-write locks and stop-the-world resizes.
- Optimization: store atomic pointers and use CAS for writes.
  - Allows updates while holding a read lock instead of a write lock.
  - But requires more advanced reclamation strategies (see next slide).
- Even better: fully lock-free hash table with chained buckets and cooperative resizes.

Safe Memory Reclamation

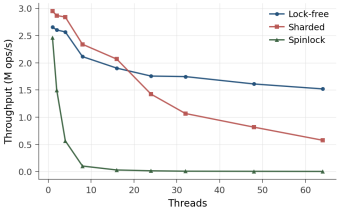
- Approaches #2 and #3 require synchronization to ensure that a value is not freed between retrieving the pointer and incrementing the ref-count.
- Traditional solution is *epoch-based reclamation*.
  - Threads announce a current "epoch"; values deleted in epoch t are not freed until all threads reach epoch $t + 1$.
  - Single memory barrier per read, but time-to-free is unbounded.
- Alternative solution: *hazard pointers*.
  - Threads announce the specific pointers they are reading (i.e., the current hash table and value); when freeing a batch of values, must check other threads' hazard lists first.
  - Much heavier and harder to implement, but stalled threads block at most two frees.

Benchmarks

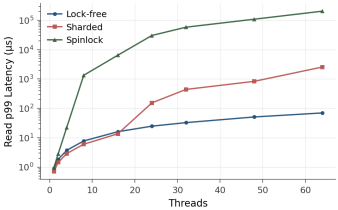
Experimental Setup

- Workload: Zipfian ($\theta = 0.99$) over 4M keys, 50% read/write, 16 threads.
  - Sweep each of these parameters, fix the others.
  - Workers pinned to a specific core (except TLB benchmark!).
  - Only microbenchmarks for now; will compare with Lightning on YCSB.
- Measurement: 5 s timed window, 2 s warmup, sampled per-op `rdtsc` (1-of-128 ops).
- Hardware: AMD EPYC 7452 (32 cores @ 2.35 GHz), 125 GB memory, 128 MB L3.

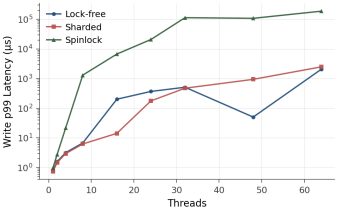
Hash Tables: Concurrency



Throughput vs. threads



Read p99 latency vs. threads



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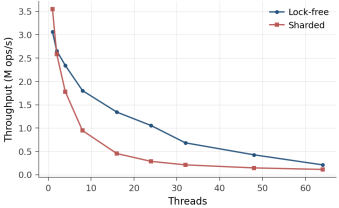
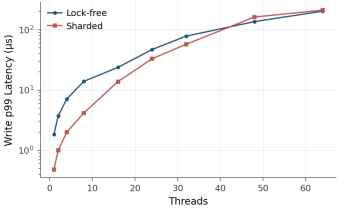


Fill throughput vs. threads



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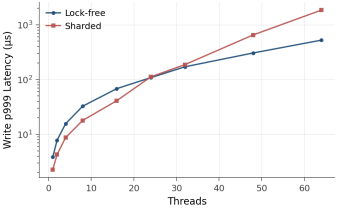
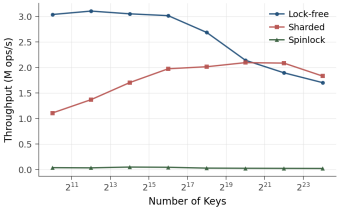
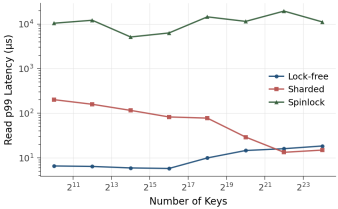


Fill p999 latency vs. threads

Hash Tables: Working Set



Throughput vs. working set



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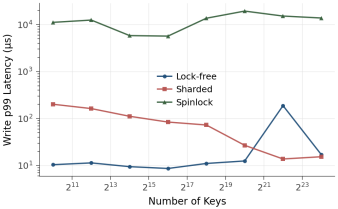

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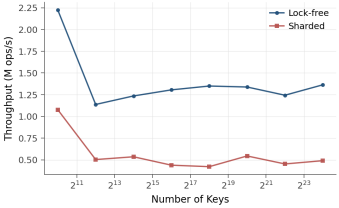

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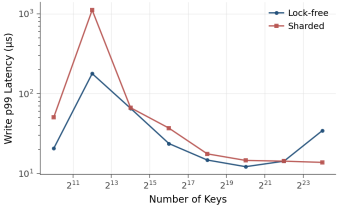

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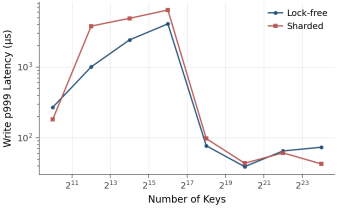
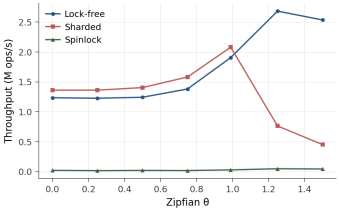
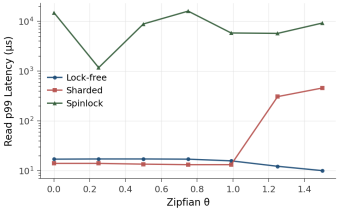


Fill p999 latency vs. working set

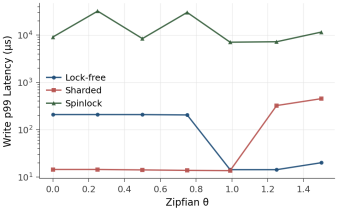
Hash Tables: Zipfian θ



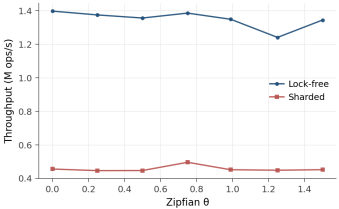
Throughput vs. θ



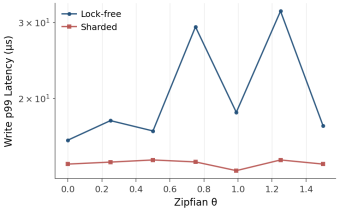
Read p99 latency vs. θ



Write p99 latency vs. θ



Fill throughput vs. θ



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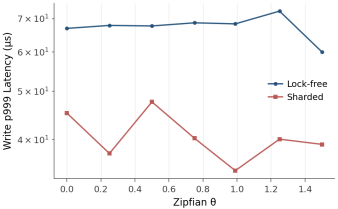
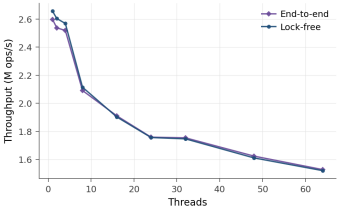
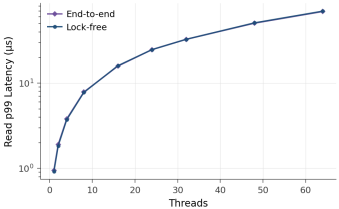


Fill p999 latency vs. θ

Client Library Overhead



Throughput vs. threads



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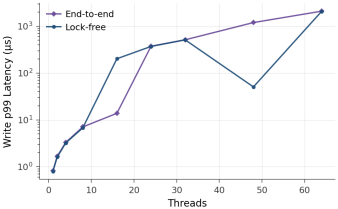
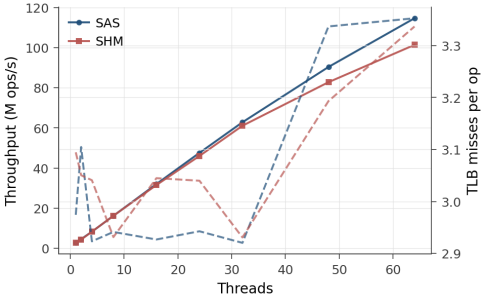
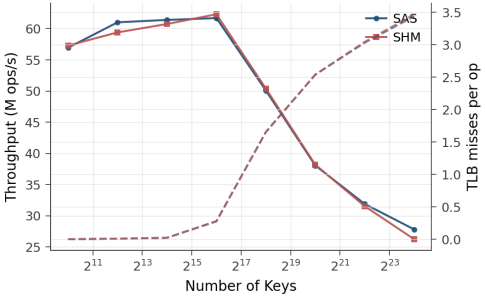


Write p99 latency vs. threads

Shared Memory Overhead



Throughput + TLB misses vs. threads



Throughput + TLB misses vs. working set

Questions?